

Khadar Chittor

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PROFESSIONAL EXPERIENCE

California State University

Graduate Student Lead Developer

California, United States · Aug 2025 - May 2026

- Optimized request handling across Python services and REST API integrations for applications, increasing backend throughput by 35%.
- Standardized Docker-based AWS deployments, reducing release issues by 25% and cutting deployment setup time from 45 to 15 minutes.
- Engineered ingestion and normalization pipelines processing 50K+ operational records per month, reducing manual data preparation by 40%.
- Implemented structured logging and failure diagnostics for backend issues, reducing production support escalations from 8 to 5 per week.
- Led a 4-member student engineering team, reviewing Python and REST API changes, resolving backend defects and coordinating Docker/AWS releases that shortened release cycles by 30%.

Graduate Student Associate Developer

California, United States · Jun 2024 - Aug 2025

- Engineered Python, C++, and PostgreSQL data-processing workflows handling 10K+ records per run, reducing batch-processing time by 30%.
- Optimized PostgreSQL reporting queries, reducing slow-query runtime from 12 seconds to 9.5 seconds.
- Automated recurring validation and reporting workflows with Python and SQL, saving 6 hours of manual operational work per week.

HCLTech

Software Engineer

Oct 2022 - Dec 2023

- Engineered Django and PostgreSQL backend services handling 100K+ transactions, implementing REST APIs for authentication, validation, persistence and error handling.
- Optimized PostgreSQL queries and refactored API endpoints, reducing average response time from 650 ms to 420 ms.
- Automated Docker-based release workflows with GitHub Actions, cutting manual deployment effort from 6 to 3 hours per sprint.
- Led a 5-member backend engineering team, reviewed Django services and PostgreSQL queries, resolved API and database blockers and coordinated QA validation through production release.

PROJECTS

AgentGate - Production Access Runtime for AI Agents

- Engineered a policy-governed runtime enabling AI agents to execute scoped actions across 10+ external systems without persistent credentials.
- Designed a seven-stage execution pipeline covering session validation, blast-radius constraints, rollback checks, policy evaluation, approval gates, connector execution and audit logging.
- Implemented session-scoped controls limiting each session to five actions and one high-risk action, with 30 to 60 minute expirations, resource allowlists and approval thresholds to constrain agent blast radius.
- Deployed the runtime with FastAPI, Next.js, PostgreSQL, Railway, and Vercel, integrating MCP-compatible agents with session-scoped authorization and execution auditing.

SCAI - Agentic Browser Automation Platform

- Engineered an agentic browser-automation platform with FastAPI, Next.js, and a Chrome extension, exposing 12 APIs for task creation, agent execution, workflow state and extension communication.
- Implemented Redis-backed orchestration with WebSocket event streaming, retries and asynchronous execution, supporting 10 concurrent sessions and improving multi-step task reliability from 72% to 89%.
- Reduced multi-step browser workflow time from 6 minutes to 2 minutes by converting manual browser sequences into reusable agent workflows.

Gen-Aistro - AI RAG Platform for NASA Space Biology Research

- Engineered an NLP/RAG pipeline over 600+ NASA space-biology papers, parsing, chunking, embedding, and indexing the corpus into 12K+ searchable chunks for semantic retrieval, summarization and question answering.
- Implemented multi-format ingestion for user-provided datasets across five document formats, feeding external research data into the same retrieval pipeline as the NASA corpus.
- Delivered a natural-language retrieval interface that reduced research discovery time from 20 minutes of manual paper review to under 2 minutes across 150+ demo and evaluation queries.

SKILLS

- **Languages:** Python, TypeScript, JavaScript, C++, SQL, Bash
- **Full Stack & Testing:** FastAPI, Django, Node.js, React, Next.js, REST APIs, gRPC, asyncio, WebSockets, OpenAPI, OAuth 2.0, JWT, pytest, Playwright, Unit & Integration Testing, Flask, Tailwind CSS, Vite, HTML5, CSS3
- **Data & Databases:** PostgreSQL, MySQL, NoSQL (MongoDB, DynamoDB), Redis, Elasticsearch, Supabase, ETL/ELT
- **Cloud & Infra:** AWS (EC2, Lambda, S3, CloudWatch, IAM, ECS, EKS, RDS, VPC), Docker, Kubernetes, CI/CD, GitHub Actions, Nginx, Linux, Git
- **AI/LLM:** Agentic AI Systems, RAG, MCP, Generative AI, Semantic Search, Embeddings, Vector Databases (Pinecone), NLP, PyTorch, LangChain, LlamaIndex, Hugging Face Transformers, Amazon Bedrock, SageMaker, MLOps, LLM APIs, Prompt Engineering

EDUCATION

California State University, Fullerton

Master of Science, Computer Science

Jan 2024 - May 2026

California, United States

Anna University, Chennai

Bachelor of Engineering, Computer Science and Engineering

Aug 2018 - Apr 2022

Chennai, India

CERTIFICATIONS & ACHIEVEMENTS

- AWS Certified Machine Learning Engineer - Associate, AWS Certified Solutions Architect - Associate
- Microsoft Certified: Azure AI Engineer Associate, CompTIA Security+, NVIDIA Generative AI & LLMs, Google Generative AI Leader
- Hackathons: 2x Winner, 4x Finalist, including NASA Space Apps Challenge, AWS YouthTech Hackathon, and Cal Hacks 12.0